

Experiment No.

Aim: To separate the constant losses of a DC machine and plot the following graph:

- Hysteresis v/s Speed
- Eddy current v/s Speed
- Mechanical loss v/s Speed

Required Instruments:

Sl.no.	Name	Type	Range	Quantity
1	Voltmeter	MC	0-300 V	1 no.
2	Ammeter	MC	0-5 A	1 no.
3	Ammeter	MC	0-2 A	1 no.
4	Rheostat	Single Tube	450 ohm, 5 A	1 no.
5	Rheostat	Single Tube	270 ohm, 1.7 A	1 no.
6	Tachometer	Digital	0-2000 RPM	1 no.

Theory:

The constant losses in DC machines are hysteresis loss, eddy current loss, friction loss, winding loss. If speed of machine is N , then

$$\text{Friction loss} = AN + BN^2$$

Hysteresis loss = CN , if excitation is kept constant.

Eddy current loss = DN^2 for constant excitation,

Where A, B, C, D are constants.

$$P = \text{Total constant loss} = \text{Input to armature circuit at no load} - \text{armature copper losses at no load}$$

Then,

$$P = AN + BN^2 + CN + DN^2$$

$$\frac{P}{N} = A + BN + C + DN$$

$$\frac{P}{N} = (A + C) + (B + D)N \text{ ----- (1)}$$

The above equation denotes straight line.

To determine all component A, B, C, D the experiment is performed with reduced excitation.

The friction losses does not changes, but hysteresis and eddy current loss will reduce and constant C & D will change to C' and D'.

The equation at the reduced excitation will be

$$\frac{P}{N} = (A + C') + (B + D')N \text{ ----- (2)}$$

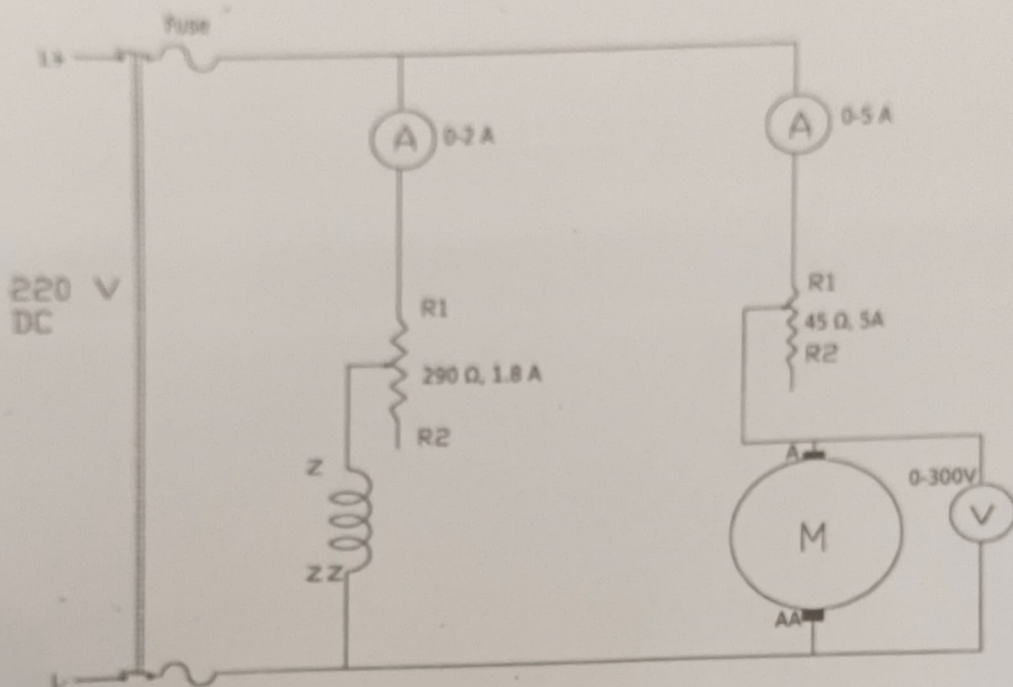
Graphically determine (A+C), (B+D), (A+C'), (B+D')

To calculate A, B, C, D the two more equations are required;

$$\begin{aligned} C/C' &= (B'/B'')^{1.6} = (I_f'/I_f'')^{1.6} \\ D/D' &= (B'/B'')^2 = (I_f'/I_f'')^2 \end{aligned}$$

Where, I_f' and I_f'' are the two values of excitation taken to perform the experiment and B' and B'' are the values of flux density corresponding to I_f' and I_f'' .

Showing these equations ABCD can be determined.



Circuit Diagram

Procedure:

1. Connect the circuit as shown in figure.
2. Adjust the armature circuit rheostat to maximum at the instant of starting.
3. Adjust the field circuit rheostat to minimum at the instant of starting.
4. Switch on the DC supply.
5. Maintain the field current constant and record the readings of all meters including speed while cutting out the armature circuit resistance in steps.
6. Switch off the machine and repeat step 2 to 5 while adjusting the field current to a reduced value (approx. 70% of earlier value) by the rheostat in the field circuit.
7. Note the speed of the motor and record all the reading of all meters.
8. Switch off the DC supply to stop the motor.
9. Measure the resistance of armature winding by voltmeter ammeter method.

Observation Table:

$I_f = \dots\dots\dots$ A (rated value)						$I_f = \dots\dots\dots$ A (70% rated value)					
S.no	I_a Amps	V volts	N rpm	P=VI	P/N	S.no	I_a Amps	V volts	N rpm	P=VI	P/N
1											
2											
3											
4											

Graph:

- Plot E v/s I characteristics
- Plot P/N v/s N characteristics

Calculations: Friction loss: ? Hysteresis loss: ? Eddy current loss: ?

Result:

Precautions:

1. Do not touch any terminal or switch without ensuring that it is dead.
2. Never work alone in the laboratory.
3. Make sure that all the connections are right, before switching on any circuit, faulty connections may cause drawing large power resulting in damage of parts of instruments.
4. The circuit should be de-energized before changing any connections.